# Synchronous phasors and the power angle

Cheat sheet · Day 14 · Fri 6 Nov · pack M11

## Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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CH8 PHASORS & POWER ANGLE (rms, per phase, generator convention: I OUT)
E0 = V + I(R + jXs)     V = |V|∠0 ref,  I = (S/V)∠phi
  phi = -acos(pf) LAGGING,  +acos(pf) LEADING        |I| = P/(V*pf)
  delta = angle(E0): >0 generating, <0 motoring
MOTOR: I_in = -I  ->  E0 = V - jXs*I_in   (same answer, delta < 0)
P = (V*E0/Xs) sin(delta)  pu = 3-phase pu (x S_base -> W)
  SI: phase V -> x3 ;  line V (VL*EL/Xs) -> already 3-phase
Pmax = V*E0/Xs at delta = 90 deg (keep OLD E0); practical limit ~75 deg
Q_gen = (V*E0*cos(delta) - V^2)/Xs  [check only; + = delivers Q]
Over-excited -> GENERATES Q (gen: lagging out; motor: leading in)
Under-excited -> ABSORBS Q (gen: leading out, big delta, low Pmax)
I_F = E0(pu) x I_F(E0 = 1 pu);  E0_line = E0(pu) x V_base,L
V-curve (const P): |I| min at pf = 1; either side |I| up (S = sqrt(P^2+Q^2))
CAPABILITY: |Q|max = sqrt(S_rated^2 - P_E^2), P_E = P_M/eta
PFC: sum P, sum Q (lag +, lead -) -> pf = cos(atan(Qtot/Ptot)), state lag/lead
Motor E for PFC: S_m = (P_E + jQ)/S_rated, I_m = conj(S_m/V), E0 = V - jXs*I_m
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